Surgical approach and outcomes in adults with anomalous aortic origin of coronary arteries at a reference center: Outcomes of proximal coronary surgery

Check for updates

Maroun Yammine, MD, Sophia Hsien, MD, Patricia Moscicki, MD, Yun Zhang, PhD, Marlon Rosenbaum, MD, Andrew J. Einstein, MD, PhD, Jonathan Kochav, MD, Alexandra Channing, MD, Edward Buratto, MD, PhD, Andrew Goldstone, MD, PhD, David Kalfa, MD, PhD, Emile Bacha, MD, and Kanwal M. Farooqi, MD

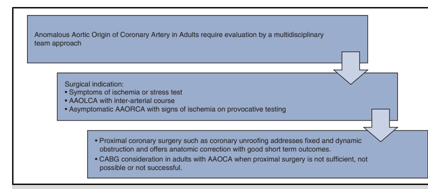
ABSTRACT

Background: Considerations in the management of anomalous aortic origin of a coronary artery (AAOCA) in adults differ from those in the pediatric population owing to the difference in risk profile. In adults for whom surgery is indicated, data on surgical outcomes can help guide decision making.

Methods: Between January 2006 and January 2023, adults who underwent surgery for AAOCA were identified from our retrospective registry that includes medically and surgically managed patients of all ages. We reviewed the preoperative and operative characteristics and in-hospital and 30-day follow-up data for the surgical adult population.

Results: A total of 316 patients with AAOCA were identified, 123 of whom (38.9%) were adults, of whom 54 (43.9%) underwent surgery. The median age of the operative adult cohort was 46 years (interquartile range, 35-52 years), and 51.9% (n = 28) were female. Presentation was because of symptoms in 85% (n = 46), including exertional chest pain in 51.9% (n = 28). Preoperative workup included cardiac computed tomography angiography in 94% (n = 51) and stress testing in 66.7% (n = 36), which was positive in 47% of these 36 patients. Anomalous left coronary was diagnosed in 35.2% of the 54 patients (n = 19), anomalous right in 63.0% (n = 34), and left coronary from noncoronary sinus in 1.9% (n = 1). Surgical approaches included unroofing in 92.6% (n = 50) with commissure resuspension in 7.4% (n = 4), and CABG in 9.2% (n = 5), as a salvage operation in 3.7% (n = 2). There was no operative mortality or stroke. New left ventricular dysfunction was severe in 1 patient (1.9%), and new aortic regurgitation was mild in 2 patients (3.7%).

Conclusions: Knowledge of the various surgical approaches is essential to providing safe treatment for adult patients with AAOCA. While unroofing should remain the mainstay approach, there remains a role for CABG when proximal surgery is not sufficient, possible, or successful. (JTCVS Open 2025;27:62-8)



Surgical approach to adults with anomalous aortic origin of the coronary artery.

CENTRAL MESSAGE

Proximal coronary surgery—namely, unroofing—is favored in the treatment for adult anomalous aortic origin of a coronary artery, owing to low associated risks. Coronary artery bypass grafting has a role when proximal surgery is not sufficient, possible, or successful.

PERSPECTIVE

This report provides guidance for all providers treating adults with anomalous aortic origin of the coronary artery and addresses the challenges faced in terms of the indication for and timing of surgery, the appropriate preoperative workup, the outcomes of proximal surgery, and the role of coronary artery bypass grafting.

See Commentator Discussion on page 79.

From the Morgan Stanley Children's Hospital, Columbia University Medical Center, New York, NY.

Read at the 104th Annual Meeting of The American Association for Thoracic Surgery, Toronto, Ontario, Canada, April 27-30, 2024.

Received for publication June 1, 2024; revisions received April 12, 2025; accepted for publication May 9, 2025; available ahead of print July 7, 2025.

Address for reprints: Kanwal M. Farooqi, MD, Morgan Stanley Children's Hospital, Columbia University Medical Center, 3959 Broadway, New York, NY 10038 (E-mail: kf2549@cumc.columbia.edu).

2666-2736

https://doi.org/10.1016/j.xjon.2025.05.011

Abbreviations and Acronyms

AAOCA	= anomalous aortic origin of the coronary artery
AAOLCA	= anomalous aortic origin of the left coronary artery
AAORCA	= anomalous aortic origin of the right coronary artery
CABG	= coronary artery bypass grafting
CHSS	= Congenital Heart Surgeons Society
IQR	= interquartile range
LVAD	= left ventricular assist device
PA	= pulmonary artery

To view the AATS Annual Meeting Webcast, see the URL next to the webcast thumbnail.

The American Association of Thoracic Surgeons expert consensus guidelines on anomalous aortic origin of coronary artery (AAOCA) surgery considered the following as class I indications for surgical intervention: any AAOCA with signs or symptoms of cardiac ischemia or anomalous aortic origin of left coronary artery (AAOLCA) with an interarterial course regardless of symptoms. It is also recommended that asymptomatic patients with anomalous aortic origin of right coronary artery (AAORCA) be evaluated for inducible ischemia and managed medically if not found (class IIA).[1,2]

More recent data and publications focus on defining high-risk anatomic features, the multiple mechanisms of ischemia which are categorized into fixed obstruction and dynamic obstruction, and how they come into play at rest and during exercise. They also emphasize the role of preoperative imaging and evaluation for stress-induced ischemia to better estimate the risk of sudden death and the reversal of ischemia following intervention.[2,3]

The challenge in determining the timing of surgery lies not only in estimating the risk of sudden death at presentation, but also in planning a safe, effective surgery with minimal morbidity. Surgery for AAOCA has been reported to have consistently low operative mortality and stroke rates but significant short- and long-term morbidity. This has been shown in data from the Congenital Heart Surgeons Society (CHSS) and multiple other single-center publications. These investigators have raised a concern for the development of aortic valve regurgitation or the need for aortic valve reintervention, neo-ostial coronary stenosis or the need for coronary intervention, and new-onset or persistent myocardial ischemia.[3-7]

The management of AAOCA varies between the adult and pediatric populations. A newly diagnosed adult may have already participated in competitive sports without incident. However, the highest mortality risk from AAOCA starts at age 10 years and extends to age 30 years.[8] To date, few reports have focused solely on adult patients with AAOCA owing to the low incidence and intervention rate.[9-11] More recently, specific anatomic features have been correlated with higher risk of ischemia in a report from the CHSS: AAOLCA diagnosis with an interarterial course, intramural course, high orifice, and slit like orifice, and AAORCA with a long intramural course.[12] This has aided the stratification of asymptomatic adults presenting with this disease.

In the pediatric population, proximal coronary surgery, such as unroofing and reimplantation, remains the mainstay of surgical treatment.[2-6] It restores normal anatomy, targets multiple (if not all) the possible reasons for ischemia, and has the potential to be a permanent solution. In adults, CABG is considered a potential option by most noncongenital surgeons. But although CABG is the most commonly performed procedure in adults and involves minimal aortic root manipulation, it does not address the primary pathology and raises other questions that remain unanswered: What is the risk of competitive flow to the bypass graft from the native coronary artery, especially if a the left interior mammary artery is used? What is the best conduit in this case, artery or vein? What is the patency rate in this young population? Should the proximal coronary arteries be ligated? There have been reports (albeit with small sample numbers) showing the poor short-term patency rate of these grafts; however, in the event of a short- or long-term proximal coronary surgery failure, CABG has potential as a salvage operation.[13]

In this study, we review our experience in the surgical approach and outcomes for AAOCA in adults at our center and report on the presentation, workup, and short operative outcomes in these patients.

METHODS

AAOCA Registry

A registry for patients with AAOCA was created retrospectively and prospectively at the Morgan Stanley Children's Hospital and Columbia University Irving Medical Center. The patients were identified initially based on a retrospective query of the electronic medical record and institutional Society of Thoracic Surgeons database for a diagnosis of AAOCA between January 2001 and January 2023. Individual charts were reviewed to confirm the diagnosis and inclusion criteria, as well as to retrieve demographic case- specific data, workup and procedure-related data, and in-hospital outcomes. Patients who had a confirmed diagnosis of AAOCA were included regardless of age or whether they received surgical or conservative management. The Institutional Review Board of the Columbia University Irving Medical Center approved the registry, the follow-up data collection, and the current study (approval AAAP3300; approved August 22, 2023). Individual consent for publication was not required. At the time of this analysis, the registry contained 316 patients of all age groups, including 123 adults and 138 who underwent surgery.

This study focuses on adults who underwent surgery for AAOCA at the Morgan Stanley Children's Hospital and Columbia University Medical Center between June 2006 and January 2023. Patients age <18 years were excluded, as were those who were medically managed or underwent surgery at another institution. Further retrospective chart review was conducted to obtain details of the surgery, inpatient postoperative events, and in-hospital and 30-day follow up. For this study, patients or providers were not contacted for long-term clinical or imaging follow-up. Categorical variables are described as percentage, and continuous variables are reported as mean with standard deviation or median and interquartile range (IQR). The primary outcome of interest was perioperative in-hospital and 30-day mortality. The secondary outcomes of interest were the perioperative risks for adverse events and need for reintervention during the same index hospitalization or the 30 days following the index operation.

RESULTS

Patient Characteristics and Presentation

Fifty-four adults had undergone surgery between June 2006 and January 2023 at the Columbia University Medical Center and the Morgan Stanley Children's Hospital. The median age was 46 years (IQR, 35.5-52.5 years), the age range was 18 to 75 years, and 51.9% (n = 28/54) were female (Table 1). The most common reason for presentation was symptoms, in 85% (n = 46) of the patients. Overall, 3.7% (n = 2/54) of the patients were diagnosed after presenting for preprocedure or premedication clearance, during workup for murmur, or as asymptomatic with a positive family history. Among symptomatic patients, exertional chest pain was the most common presenting complaint (51.9%; n = 28), followed by exertional dyspnea (24.1%; n = 13), noncardiac chest pain (20.4%; n = 11), and palpitations (18.5%; n = 10). Overall, 44.4% (n = 24) of the patients had nonischemic symptoms, and 38.9% (n = 21) had multiple symptoms at the time of presentation; however, all patients who underwent surgery had either ischemic symptoms or stress-induced ischemia.

TABLE 1. Demographics and preoperative evaluation (N = 54)

Parameter	Value
Age, y, median (IQR)	46 (35.5-52.5)
Female sex, % (n)	51.9 (28)
BMI, median (IQR)	27.8 (23.7-32.2)
Presentation, % (n)	
Symptomatic referral	85.2 (46)
Preprocedure or premedication clearance	3.7 (2)
Family history/asymptomatic	3.7 (2)
Murmur	3.7 (2)
Other existing cardiac disease	3.7 (2)
Symptoms, % (n)	
Exertional chest pain	51.9 (28)
Noncardiac chest pain	20.4 (11)
Exertional syncope	3.7 (2)
Exertional dyspnea	24.1 (13)
Nonexertional dyspnea	5.5 (3)
Exertional presyncope	9.3 (5)
Palpitations	18.5 (10)
Aborted SCD	3.7 (2)
Multiple symptoms	38.9 (21)
Preoperative workup, % (n)	
Echocardiography	90.7 (49)
Cardiac CTA	94.4 (51)
Cardiac MRI	14.8 (8)
Exercise stress test	20.4 (11)
Nuclear stress test	40.7 (22)
Stress echocardiography	3.7 (2)
Catheterization	59.3 (32)
Multiple preoperative tests	94.4 (51)
LVEF <60%	9.3 (5)
Other echocardiographic findings	
Aortic stenosis (valvular and subvalvular)	7.4 (4)
VSD with aortic regurgitation	1.8 (1)
Severe tricuspid regurgitation, RV dysfunction	1.8 (1)
Wall motion abnormalities	1.8 (1)
Underwent cardiac MRI, % (n)	9.3 (5)
Exercise stress test performed, % (n)	22.2 (12)
Nuclear stress test performed, % (n)	38.9 (21)
Stress TTE performed, % (n)	5.6 (3)
Positive stress test, % (n)	29.6 (16)

IQR, Interquartile range; *BMI*, body mass index; *SCD*, sudden cardiac death; *CTA*, computed tomography angiography; *MRI*, magnetic resonance imaging; *LVEF*, left ventricular ejection fraction; *VSD*, ventricular septal defect; *RV*, right ventricular; *TTE*, transthoracic echocardiography.

Diagnosis and Workup

Preoperatively, 90.7% (n = 49) of the patients had undergone transthoracic echocardiography, and 94% (n = 51) had undergone cardiac computed tomography angiography. Some form of stress testing was performed in 66.7% (n = 36) of the patients, and tests were positive in 44.4% (16/36). Catheter angiography was performed in 59.3% (n = 32) of the patients; cardiac magnetic resonance imaging, in 14.8% (n = 8). The majority of patients underwent multiple forms of preoperative testing (94.4%; n = 51). Concomitant echocardiographic findings at the time of workup included left ventricular ejection fraction <60% in 9.3% (n = 5) among other findings, as shown in Table 1. Patients in our series were operated on because they had either ischemic symptoms or inducible ischemia on stress testing, regardless of their anatomy.

Intraoperatively, anatomy was consistent with AAOLCA in 35.2% (n = 19) of the patients, which included a single right coronary in 7 patients and a subpulmonic intraconal coronary in 6 patients. AAORCA was noted in 63.0% (n = 34), and left coronary from the noncoronary sinus was seen in 1.9% (n = 1).

Operative Characteristics

The majority of the procedures (81.5%; n = 44) were performed by a congenital surgeon as the primary surgeon or as a cosurgeon. An intramural course and a slit-like ostium were the most prevalent intraoperative findings (79.6% [n = 43] and 59.3% [n = 32], respectively) (Table 2). An intraconal or subpulmonic course was observed in 11.1% (n = 6) of the patients. The most common repair technique performed was unroofing, in 92.6% (n = 50) of patients, with commissure takedown in 14.8% (n = 8) and resuspension in 7.4% (n = 4). Unroofing was performed in combination with other procedures in 5.6% (n = 3) of cases and included reimplantation, pulmonary artery (PA) translocation, and CABG. Isolated coronary reimplantation was performed in 3.7% (n = 2) of cases. Concomitant procedures were uncommon and included aortic valve replacement in 3.7% (n = 2), for primary severe aortic stenosis in 1 patient and primary severe aortic regurgitation in 1 patient. Patients with AAOLCA and a subpulmonary intraconal course constituted 11.1% (n = 6) of the total cohort. Five of these patients underwent PA division and posterior coronary unroofing; of these, 1 patient underwent a concomitant PA translocation, and 1 underwent concomitant CABG because of an atherosclerotic lesion. The sixth patient underwent isolated 2-vessel CABG (Table 3).

CABG was performed in 9.2% (n = 5) of the patients in 2 settings, either as a primary procedure in 5.5% (n = 3) of the patients or as a salvage return to the operating room only in 3.7% (n = 2). In 2 primary CABG cases, the operation was performed when no other approach was feasible owing to high-risk anatomy or to a single coronary ostium from left sinus with a diminutive right coronary artery ostium and a short intramural segment. Whether these 2 patients were evaluated for reimplantation is unclear. In the third patient, the operation was performed to treat a distal left anterior descending coronary atherosclerotic lesion concomitantly with unroofing of an intraconal AAOLCA. One of the 2 salvage CABGs was performed following a cardiac arrest in the intensive care unit within hours after an unroofing procedure, necessitating extracorporeal membrane oxygenation and an emergent return to the operating room. Notably, the patient's left ventricular dysfunction did not recover following CABG, and he underwent LVAD placement and subsequent heart transplant. The second salvage CABG was performed 2 weeks following an unroofing procedure. The patient's symptoms returned, with evidence of recurrent ischemia. This was considered a failure of the initial proximal coronary intervention, and thus CABG was performed. In this series, the left internal mammary artery was used to bypass the left anterior descending coronary artery, and a saphenous vein graft was used to bypass any other coronary location. Proximal ligation of a coronary artery was not performed.

TABLE 2. Operative characteristics (N = 54)

Parameter	Value, % (n)
Pediatric surgeon as primary or cosurgeon	81.5 (44)
Adult surgeons only	18.5 (10)
Intraoperative diagnosis	
AAOLCA	35.2 (19)
AAORCA	63.0 (34)
LCA from nonsinus	1.9 (1)
Intraoperative features	
Intramural course	79.6 (43)
Slit-like ostium ("fish-mouth")	59.3 (32)
Interarterial course	18.5 (10)
Intraconal (subpulmonic)	11.1 (6)
Acute angle takeoff	3.7 (2)
Repair technique	
Unroofing	92.6 (50)
Commissure takedown	14.8 (8)
Commissure resuspension	7.4 (4)
Reimplantation	3.7 (2)
Pulmonary artery translocation	1.9 (1)
CABG	5.6 (3)
Mixed repair techniques	5.6 (3)
Concomitant procedures	18.5 (10)
Aortic valve replacement	3.7 (2)
VSD	1.9 (1)
Myectomy	3.7 (2)
Valve sparing root replacement	1.9 (1)
Mitral valve repair	1.9 (1)
Tricuspid valve repair	1.9 (1)
PFO closure	1.9 (1)

AAOLCA, Anomalous aortic origin of left coronary artery; *AAORCA*, anomalous aortic origin of right coronary artery; *LCA*, left coronary artery; *CABG*, coronary artery bypass grafting; *VSD*, ventricular septal defect; *PFO*, patent foramen ovale.

Results

There was no operative mortality and no incidence of stroke in any of our patients (Table 4). New postoperative

TABLE 3. Surgical techniques by anatomic lesion

Anatomic lesion	Surgical technique
AAOLCA*	Unroofing (n = 11)
	CABG (n = 1)
	Reimplantation (n = 1)
Subpulmonic	Unroofing
	Unroofing and CABG (n = 1)
	Unroofing and PA translocation (n = 1)
LCA from noncoronary artery	Unroofing (n = 1)
AAORCA	Unroofing (n = 30)
	Unroofing then salvage CABG (n = 1)
	Unroofing then reimplantation (n = 1)
	Isolated CABG (n = 2)

AAOLCA, Anomalous aortic origin of the left coronary artery from the right coronary sinus; *CABG*, coronary artery bypass grafting; *PA*, pulmonary artery; *LCA*, left coronary artery; *AAORCA*, anomalous aortic origin of the right coronary artery. *This group does not include the AAOLCA with subpulmonic/intraconal subgroup.

TABLE 4. Outcomes

Outcome	Value
Mortality, % (n)	0.0 (0)
Stroke, % (n)	0.0 (0)
Extracorporeal membrane oxygenation, % (n)	1.9 (1)
Worsened function, % (n)	3.7 (2)
Reoperation or reintervention, % (n)	1.9 (1)
Reexploration for bleeding, % (n)	3.7 (2)
Aortic insufficiency ≤mild, % (n)	3.7 (2)
Aortic insufficiency >mild, % (n)	0.0 (0)
Pericardial effusion, % (n)	11.1 (6)
Pericarditis, % (n)	16.7 (9)
Mediastinitis, % (n)	0.0 (0)
Length of stay, d, median (IQR)	6 (5-7)

IQR, Interquartile range.

left ventricular dysfunction occurred in 3.8% (n = 2) of the patients, which was mild in 1 and severe in the other. New postoperative aortic valve regurgitation was observed in 3.8% (n = 2) and was mild in both patients. No patient required aortic valve replacement following proximal coronary surgery. The median hospital length of stay was 6 days (IQR, 5-7 days). The incidence of postoperative pericarditis requiring treatment was 16.7% (n = 9), and that of pericardial effusion requiring drainage was 11.1% (n = 6).

DISCUSSION

Our study demonstrates that proximal coronary surgery for AAOCA can be performed in adults as it has been described in the pediatric population, with no perioperative mortality or stroke. The safety of this procedure has been demonstrated in multiple reports that include all age groups with a 0% to 1% risk of mortality (Figure 1).[3-7]

Typical ischemic chest pain was seen in 51.9% of our patients, and 44% of patients had nontypical ischemic symptoms. However, patients were operated on in our series because of ischemic symptoms or inducible ischemia on stress testing, regardless of their anatomy.

The most commonly performed surgical technique in our experience was unroofing, in 93% of patients. According to the CHSS database, unroofing remains the most frequently performed procedure for AAOCA by congenital surgeons, because it achieves an appropriate anatomic position of the coronary ostium and address fixed and dynamic mechanisms of obstruction.[3-7] In adults, CABG is considered more frequently, as the most commonly performed procedure by adult cardiac surgeons (albeit for the diagnosis of coronary artery disease). In our current series, CABG was required in 5.5% of patients as a primary procedure and in 3.7% as a salvage operation. This patient population typically has a higher incidence of coronary artery disease, and as such, there remains a role for CABG in adults with AAOCA in circumstances where proximal surgery is not sufficient, not possible, or not successful. The report by Fedoruk and colleagues clearly demonstrates the high risk of short-term right internal mammary artery failure in this setting and discusses considerations in appropriate conduit selection and evaluation for potential ligation of the proximal coronary artery.[13] We therefore advocate for a discussion and potential collaboration with adult surgeons who perform these surgeries frequently.

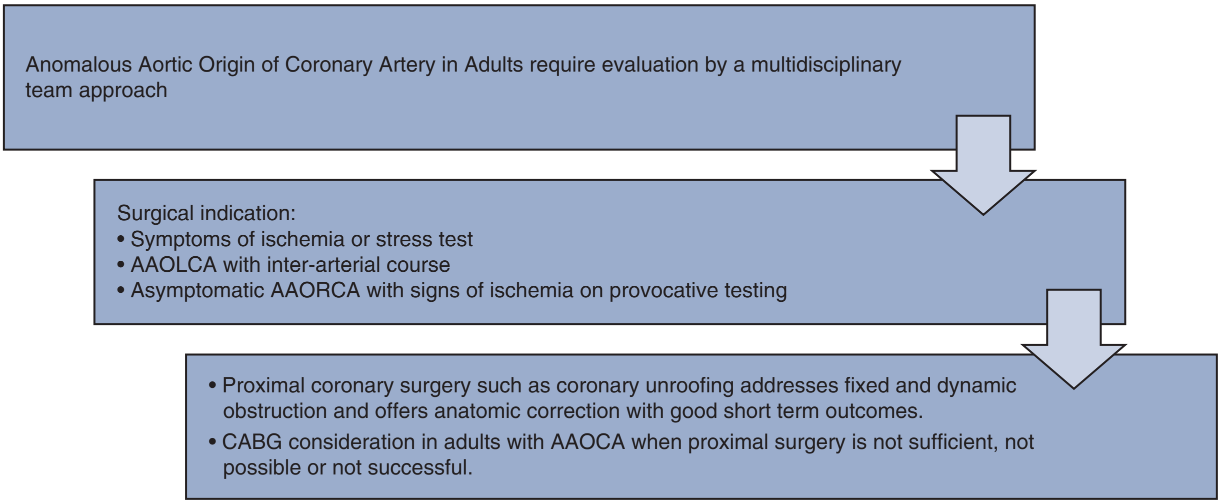


FIGURE 1. The typical approach to adults with anomalous aortic origin of the coronary artery, which requires multidisciplinary team evaluation, identification of the indication for surgery, and then consideration for proximal coronary surgery or coronary artery bypass grafting depending on a case-by-case evaluation. *AAOLCA*, Anomalous aortic origin of left coronary artery; *AAORCA*, anomalous aortic origin of right coronary artery; *CABG*, coronary artery bypass grafting.

Our study echoes the results of previously reported surgical series focused on the morbidity following this procedure. In our cohort, 3 patients (5.5%) had some degree of left ventricular dysfunction and 1 patient required left ventricular support with eventual heart transplant after a cardiac arrest. This rate was similar to the reported 4% rate of new-onset of ischemia or wall motion abnormality postoperatively in some previous reports.[3,6] The exact mechanism of new-onset ischemia remains unclear and is likely multifactorial. Concomitant coronary artery atherosclerotic lesions, insufficient unroofing to an anatomically appropriate location, and intraoperative suboptimal protection are several of the possible etiologies. In our series, 2 of the patients who had left ventricular dysfunction had subpulmonic/intraconal AAOLCA. One of them had mild left ventricular dysfunction at the time of discharge, and the other had moderate to severe global left ventricular dysfunction following PA division and unroofing. Further investigation into this phenomenon will facilitate the development of techniques to prevent this morbidity.

Jegatheeswaran and colleagues[3] reported a freedom from coronary-related reoperations of 95% at 7 years, as well as 88% freedom from greater than mild AI without commissural manipulation, compared to 77% with commissural manipulation at a 3-year follow-up. In our series, 1 patient required a coronary-related reoperation early following his unroofing. Two patients (3.7%) experienced new-onset postoperative aortic regurgitation, which was mild in both cases.

Destabilizing the aortic valve at the level of the commissure remains a major concern for proximal coronary surgery for AAOCA; however, multiple techniques have been developed to mitigate this risk. These techniques include commissure takedown and resuspension, as well as a more cephalad opening of the common aortocoronary wall at the level of the commissure. The commissure is then assessed for resuspension following complete unroofing, taking into consideration any impact on the neo-ostium. This later variation of commissure takedown technique is our currently preferred approach, whereas others have adopted a reimplantation technique that minimizes manipulation of the commissure.[14]

Interestingly in our series, the incidence of postoperative pericarditis was 16.7% and that of pericardial effusion was 11.1%. These patients were managed primarily conservatively with anti-inflammatory drugs and follow-up echocardiography. Pericardiocentesis and a pericardial window were performed in patients with extensive or symptomatic pericardial effusion.

Finally, our AAOCA registry was created in an effort to standardize the evaluation and management of this disease in children and adults and track the progress of the multidisciplinary team efforts in managing these patients. As we continue to recruit patients prospectively into the registry, case by case discussions are held among imaging, clinical and surgical specialists in congenital, adult congenital, and adult cardiac disease to choose the optimal plan for each individual patient. This taskforce also has established a management algorithm for AAOCA patients who have and those who have not undergone surgical intervention. Other referral centers also have established congenital coronary artery programs that have developed algorithms to help streamline the evaluation and improve the outcomes of surgical interventions.[14]

LIMITATIONS

This study is a retrospective review of surgical cases from our AAOCA registry and will inherently carry the limitations of such a study design. Long-term follow-up is needed to determine the efficacy of this procedure in terms of symptom control and success in resuming exercise and competitive sports, as well as to assess aortic valve competence. Postoperative imaging to rule in or rule out new potential mechanisms for recurrent ischemia was not available.

CONCLUSIONS

Adults with AAOCA with signs or symptoms of ischemia or high-risk anatomic features should be considered for a surgical intervention involving proximal coronary surgery, as it carries low operative mortality. Patients should be counseled about the morbidity from this surgery. There remains a role for CABG for patients in whom proximal surgery is not sufficient, not successful, or not possible.

Webcast

You can watch a Webcast of this AATS meeting presentation by going to: http://aats.org/resources/surgical-approach-and-outcomes-7135.

Conflict of Interest Statement

The authors reported no conflicts of interest.

The *Journal* policy requires editors and reviewers to disclose conflicts of interest and to decline handling or reviewing manuscripts for which they may have a conflict of interest. The editors and reviewers of this article have no conflicts of interest.

References

1. Brothers JA, Frommelt MA, Jaquiss RDB, Myerburg RJ, Fraser CD Jr, Tweddell JS. Expert consensus guidelines: anomalous aortic origin of a coronary artery. *J Thorac Cardiovasc Surg*. 2017;153(6):1440-1457.

2. Gaudino M, Di Franco A, Arbustini E, et al. Management of adults with anomalous aortic origin of the coronary arteries: state-of-the-art review. *J Am Coll Cardiol*. 2023;82(21):2034-2053.
3. Jegatheeswaran A, Devlin PJ, Williams WG, et al. Outcomes following anomalous aortic origin of a coronary artery repair: a Congenital Heart Surgeons' Society study. *J Thorac Cardiovasc Surg*. 2020;160(3):757-771.e5.
4. Fabozzo A, DiOrio M, Newburger JW, et al. Anomalous aortic origin of coronary arteries: a single-center experience. *Semin Thorac Cardiovasc Surg*. 2016;28(4):791-800.
5. Sachdeva S, Frommelt MA, Mitchell ME, Tweddell JS, Frommelt PC. Surgical unroofing of intramural anomalous aortic origin of a coronary artery in pediatric patients: single-center perspective. *J Thorac Cardiovasc Surg*. 2018;155(4):1760-1768.
6. Nees SN, Flyer JN, Chelliah A, et al. Patients with anomalous aortic origin of the coronary artery remain at risk after surgical repair. *J Thorac Cardiovasc Surg*. 2018;155(6):2554-2564.e3.
7. Mery CM, De León LE, Molossi S, et al. Outcomes of surgical intervention for anomalous aortic origin of a coronary artery: a large contemporary prospective cohort study. *J Thorac Cardiovasc Surg*. 2018;155(1):305-319.e4.
8. Taylor AJ, Byers JP, Cheitlin MD, Virmani R. Anomalous right or left coronary artery from the contralateral coronary sinus: "high risk" abnormalities in the initial coronary artery course and heterogeneous clinical outcomes. *Am Heart J*. 1997;133(4):428-435.
9. Warner ED, Gulati A, Halpern E, et al. Clinical outcomes in adult patients with an anomalous right coronary artery from the left sinus of Valsalva. *Am J Cardiol*. 2023;204:122-129.
10. Davis JA, Cecchin F, Jones TK, Portman MA. Major coronary artery anomalies in a pediatric population: incidence and clinical importance. *J Am Col Cardiol*. 2001;37(2):593-597.
11. Eckart RE, Scoville SL, Campbell CL, et al. Sudden death in young adults: a 25 -year review of autopsies in military recruits. *Ann Intern Med*. 2004;141(11):829-834.
12. Jegatheeswaran A, Devlin PJ, McCrindle BW, et al. Features associated with myocardial ischemia in anomalous aortic origin if a coronary artery: a Congenital Heart Surgeons' Society study. *J Thorac Cardiovasc Surg*. 2019;158(3):822-834.e3.
13. Fedoruk LM, Kern JA, Peeler BB, Kron IL. Anomalous origin of the right coronary artery: right internal thoracic artery to right coronary artery bypass is not the answer. *J Thorac Cardiovasc Surg*. 2007;133(2):456-460.
14. Kohlsaat K, Gauvreau K, Beroukhim R, Newburger J, Quinonez L, Nathan M. Trends in surgical management of anomalous aortic origin of the coronary artery over 2 decades. *JTCVS Open*. 2023;16:757-770.

Key Words: congenital heart disease, anomalous coronary artery